

1 Solution — GIAM 1.3

Problem 1

1.1 Problem

How many quantifiers (and what sorts) are in the following sentence?

“Everybody has some friend that thinks they know everything about a sport.”

1.2 Answer

There are **four** quantifiers — **two universal** ($\forall$) and **two existential** ($\exists$):

Word in the sentence	Quantifier	Type
“Everybody”	$\forall$	universal (“for all”)
“some friend”	$\exists$	existential (“there is”)
“everything”	$\forall$	universal (“for all”)
“a sport”	$\exists$	existential (“there is”)

Table 1.1

Read left to right, they alternate: $\forall \exists \forall \exists$.

1.3 Identifying Each One

- **“Everybody”** — ranges over *all* people: a universal quantifier, $\forall$. (“Every,” “all,” “each,” “everybody” are the tell-tale universal words.)
- **“some friend”** — asserts the *existence* of at least one friend: an existential quantifier, $\exists$. (“Some,” “a,” “there is,” “at least one” signal existence. The textbook even italicised *some* to flag it.)

- “**everything**” — “knows *everything*” means *all* the relevant facts: a universal quantifier, $\forall$, ranging over facts.
- “**a sport**” — “about *a* sport” asserts the existence of a sport: an existential quantifier, $\exists$.

1.4 The Four Quantifiers at a Glance

Four quantifiers: universal ($\forall$, blue) and existential ($\exists$, orange), alternating $\forall \exists \forall \exists$.

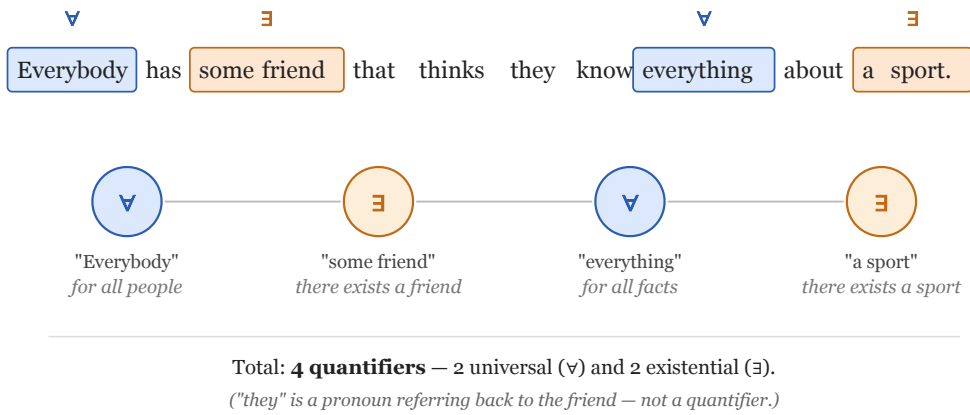


Figure 1.1: Each quantifier phrase is boxed and tagged: ‘Everybody’ ($\forall$), ‘some friend’ ($\exists$), ‘everything’ ($\forall$), ‘a sport’ ($\exists$). The pattern across the sentence is $\forall \exists \forall \exists$. The word ‘they’ is a pronoun, not a quantifier.

1.5 Symbolic Form

Spelling the sentence out in logic makes the four quantifiers explicit. Let the universe of *people* be P , write $\text{Fr}(x, y)$ for " y is a friend of x ", $\text{Sp}(s)$ for " s is a sport", $\text{Ab}(f, s)$ for " f is a fact about s ", and $\text{Kn}(y, f)$ for " y knows f ". One faithful rendering is:

$$\forall x \exists y \left[\text{Fr}(x, y) \wedge \exists s (\text{Sp}(s) \wedge \forall f (\text{Ab}(f, s) \rightarrow \text{Kn}(y, f))) \right].$$

The four quantifiers $\forall x, \exists y, \exists s, \forall f$ are exactly the four English phrases “everybody,” “some friend,” “a sport,” “everything.”

1.6 Remark — The Alternating $\forall \exists \forall \exists$ Pattern

Reading the quantifier words in the order they *appear* in the sentence gives

$$\underbrace{\forall}_{\text{everybody}} \quad \underbrace{\exists}_{\text{some friend}} \quad \underbrace{\forall}_{\text{everything}} \quad \underbrace{\exists}_{\text{a sport}},$$

a perfectly alternating $\forall\exists\forall\exists$. This is almost certainly the point of the exercise: it is the *same* four-quantifier $\forall\exists\forall\exists$ shape that the very next exercise (Exercise 4) asks the reader to *build* from scratch. Sentences with several alternating quantifiers are exactly the ones the section warns are easy to misread.

1.7 Remark — Scope: “Knows Everything About a Sport”

The phrase “knows everything about a sport” hides a subtlety the bare count glosses over. Idiomatically it means “*is an expert on some particular sport*” — i.e. **there is a sport** such that the friend knows **every** fact about it:

$$\exists s (\text{Sp}(s) \wedge \forall f (\text{Ab}(f, s) \rightarrow \text{Kn}(y, f))).$$

Here the existential “a sport” actually takes *wider* scope than the universal “everything,” even though “everything” appears *first* in the English. So the surface reading order ($\dots\forall\text{everything}\exists\text{a sport}$) and the logical nesting ($\dots\exists\text{a sport}\forall\text{fact}$) disagree in order. Either way the *count* is unchanged — two universals and two existentials — but the example shows why “just count the words” is only the first step; pinning the *scopes* is where the real care lies.

1.8 Remark — What Is *Not* a Quantifier

It is worth being explicit about the words that look grammatically important but carry no quantification:

- “**they**” — a *pronoun*, referring back to “some friend.” In logic it becomes a *bound variable* (the y already introduced by $\exists y$), not a new quantifier.
- “**has**,” “**thinks**,” “**know**” — verbs (predicates/relations), not quantifiers.
- “**that**” — a relative pronoun introducing the clause “that thinks...”; structural glue, not a quantifier.

Only the four determiner/quantifier words — *everybody*, *some*, *everything*, *a* — introduce quantification.

1.9 Remark — Why the Order of Quantifiers Matters

The section stresses that swapping quantifiers can change a sentence’s meaning entirely. Compare the textbook’s cafeteria example:

- “For *every* student, there is *some* food they like” ($\forall\exists$) — each student has a (possibly different) favourite.
- “There is *some* food that *every* student likes” ($\exists\forall$) — one single food pleases everyone.

These are genuinely different claims. The same sensitivity lives in our sentence: “Everybody has *some* friend...” ($\forall\exists$: each person’s expert friend may differ) says something far weaker than “There is *some* friend that everybody has...” ($\exists\forall$: one universally-shared friend). When a sentence stacks four alternating quantifiers, this ordering effect compounds — which is exactly why counting and *typing* them, as this exercise asks, is the necessary first discipline before reasoning about meaning.

1.10 Remark — The Universe of Discourse for Each Quantifier

Each quantifier silently ranges over a different *universe of discourse*, and naming them is good hygiene (the section’s recurring theme):

Quantifier	Ranges over
“everybody” ($\forall$)	people
“some friend” ($\exists$)	people (a friend <i>is</i> a person)
“everything” ($\forall$)	facts about the chosen sport
“a sport” ($\exists$)	sports

Table 1.2

Leaving these implicit is harmless in casual English, but in a careful proof the universe of each quantifier must be declared — otherwise statements like “everything” are ambiguous (everything in the world? every fact? every fact *about that sport?*). Making the universes explicit is precisely the “scary notation” discipline this section is introducing.